CYBERSECURITY LAB REPORT

Network Intrusion Detection Using Snort IDS/IPS

Packet Capture and ICMP Ping Alert Analysis

Submitted by	Md. Jahid Matubber
Platform	Kali Linux
Security tool	Snort IDS/IPS

Abstract

This report documents a network monitoring exercise using Snort, an open-source network intrusion detection and prevention system. The supplied evidence shows a packet capture containing local network traffic and a Snort startup sequence in which the configuration and rule files are loaded. Snort is then shown processing packets on the eth0 interface in passive mode and generating repeated ICMP ping detection messages for traffic between 192.168.10.129 and 192.168.10.2. The report describes only the activity visible in the two-page evidence file and does not claim that packet blocking or inline prevention was performed.

1. Introduction

Network traffic monitoring helps administrators identify suspicious communication and possible reconnaissance activity. Snort can inspect packets and compare them with configured rules to produce alerts when traffic matches a detection condition. This lab focuses on starting Snort, loading its configuration and rules, monitoring an interface, and observing ICMP ping alerts.

2. Objectives

- Observe network packets through a packet-capture interface.
- Load the Snort configuration and rule files successfully.
- Start packet processing on the eth0 network interface.
- Observe Snort alerts for ICMP ping traffic.
- Explain the difference between passive detection and inline prevention.

3. Lab Environment and Scope

Component	Evidence-based information
Operating environment	Kali Linux terminal and packet-capture application visible in screenshots
IDS/IPS software	Snort; exact version is not legible/provided
Network interface	eth0
Observed IP addresses	192.168.10.129 and 192.168.10.2
DAQ mode	Passive, as reported by Snort output
Evidence	Two pages containing packet-capture and Snort terminal screenshots

4. Methodology and Procedure

4.1 Packet capture observation

Page 1 contains a packet-capture screenshot. The visible traffic includes repeated ARP broadcast frames and several IPv6-related packets, including ICMPv6 and mDNS. The lower portion also shows local network traffic such as DHCP and other discovery/control messages. This establishes that network packets were being observed, but the screenshot alone does not identify every endpoint or prove malicious activity.

4.2 Snort initialization and rule loading

Page 2 shows Snort loading its Lua configuration from /etc/snort/snort.lua, followed by the IPS rules and the local rules file. The displayed policy statistics report one loaded policy and one enabled policy. The rule summary shows one total rule, consisting of one text rule, one option chain and one chain header. The port-rule summary shows one ICMP rule and zero TCP, UDP and IP port rules in the displayed configuration.

4.3 Packet processing and alert observation

The terminal output states that the PCAP DAQ is configured to passive mode and that packet processing begins on eth0. The subsequent alert lines repeatedly report “ICMP Ping Detected” for traffic in both directions between 192.168.10.129 and 192.168.10.2. The output therefore supports that the configured ICMP detection rule matched observed ping packets.

5. Results and Observations

Area	Observed evidence	Interpretation
Packet capture	ARP broadcasts, ICMPv6, mDNS and other local network packets visible on page 1.	Packet capture/observation is demonstrated.
Configuration	Snort loads /etc/snort/snort.lua and rule files on page 2.	Configuration and rules were read by Snort.
Rule statistics	One rule loaded; one ICMP rule shown; zero TCP/UDP/IP port rules.	The active rule set shown is very small and focused on ICMP.
DAQ operation	“PCAP DAQ configured to passive” and packet processing on eth0.	Passive monitoring is evidenced; inline blocking is not.
Detection alert	Repeated “ICMP Ping Detected” messages for 192.168.10.129 and 192.168.10.2.	The ICMP rule matched the observed ping traffic.

6. Discussion

The most direct result is the generation of ICMP ping alerts. A ping is commonly used to test whether a host is reachable, and repeated ping traffic can also appear during network discovery. An alert indicates that the configured rule matched the packet; it does not, by itself, prove an attack. The two IP addresses are shown in the alert output, but the supplied evidence does not establish which address belongs to a particular user or device.

The passive DAQ mode means Snort is observing traffic rather than demonstrating inline packet dropping. Although Snort supports IPS functionality in suitable inline configurations, no successful blocking action is visible in the submitted screenshots.

7. Recommendations

- Add and validate rules for the network protocols and services that need monitoring.
- Use a controlled test to generate known ICMP traffic and compare it with the resulting alerts.
- Review alert timestamps, source/destination addresses and rule identifiers during investigations.
- Tune rules to reduce false positives while retaining useful detection coverage.
- Protect the Snort configuration and local rule files with appropriate permissions and change control.

- If prevention is required, configure and test an approved inline deployment; passive monitoring alone does not block packets.
- Keep Snort and its rules updated and regularly review logs.

8. Limitations

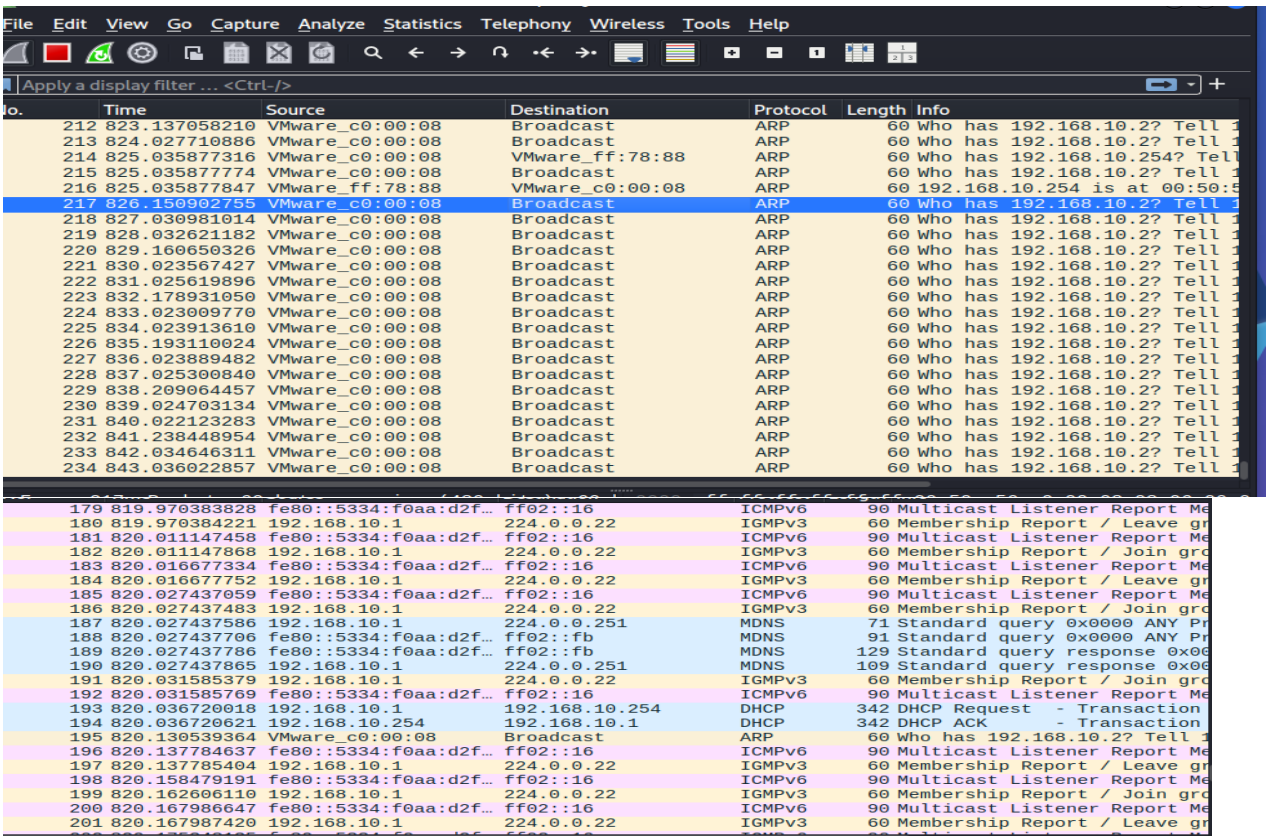
The report is based only on the two pages supplied. The exact Snort version, full rule contents, complete packet capture, command line used to start Snort, and independent endpoint identity information are not available. The evidence shows passive packet processing and ICMP alerts, but does not demonstrate packet dropping, successful prevention, or detection of a confirmed malicious intrusion.

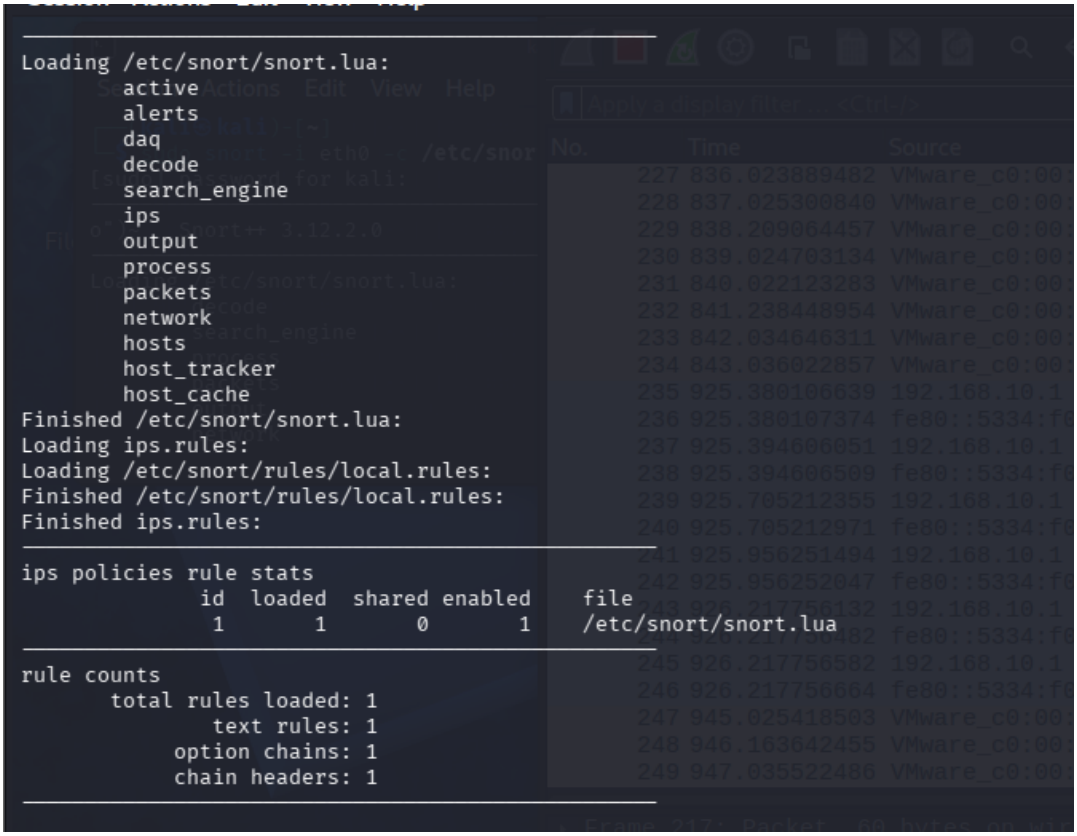
9. Conclusion

The lab evidence demonstrates successful Snort initialization, rule loading and passive packet processing on eth0. Snort generated repeated ICMP ping alerts for traffic between 192.168.10.129 and 192.168.10.2, showing that the configured ICMP rule was active and matching packets. The exercise illustrates the basic operation of a network IDS and highlights the importance of appropriate rule coverage, alert analysis and careful distinction between detection and prevention.

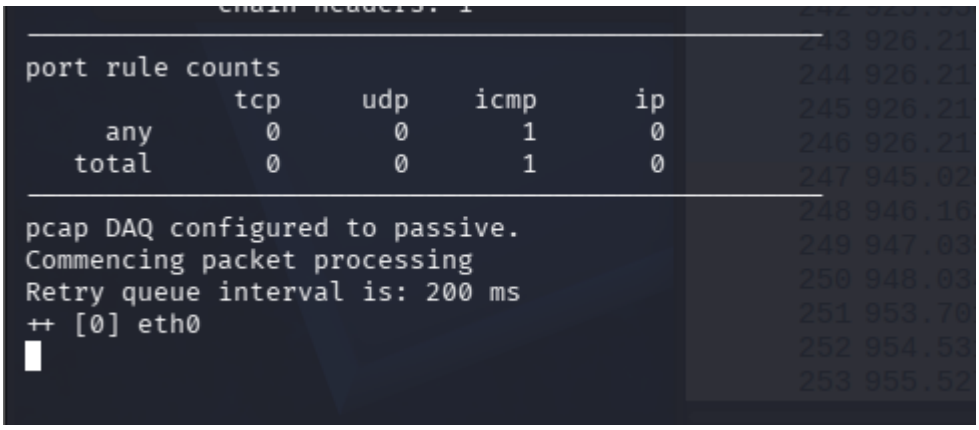
Appendix A. Original Screenshot Evidence

The original two pages from the supplied PDF are reproduced below in their original order.

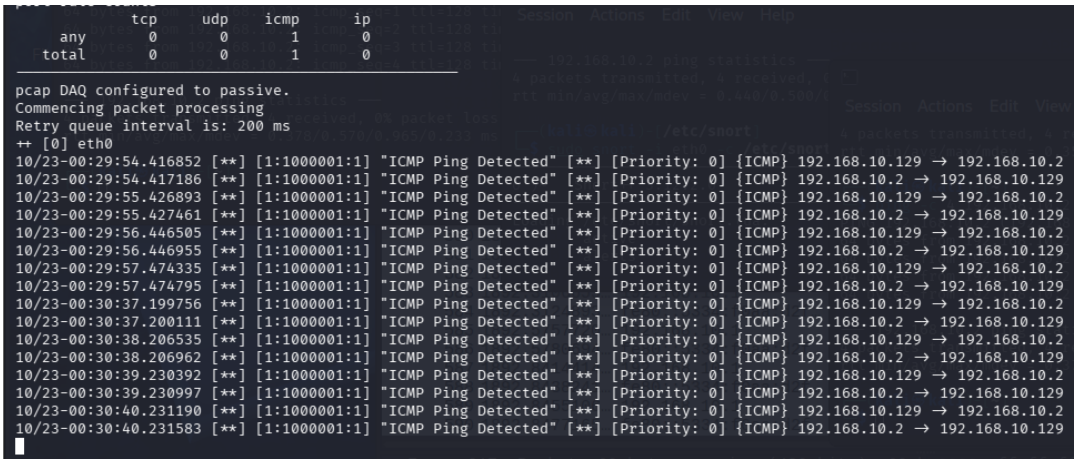




```
Loading /etc/snort/snort.lua:
        active
        alerts
        daq
        decode
        search_engine
        ips
        output
        process
        packets
        network
        hosts
        host_tracker
        host_cache
Finished /etc/snort/snort.lua:
Loading ips.rules:
Loading /etc/snort/rules/local.rules:
Finished /etc/snort/rules/local.rules:
Finished ips.rules:
------------------------------------------------------------
ips policies rule stats
              id  loaded  shared enabled    file
               1       1       0       1    /etc/snort/snort.lua
------------------------------------------------------------
rule counts
       total rules loaded: 1
                text rules: 1
              option chains: 1
              chain headers: 1
------------------------------------------------------------
```



```
------------------------------------------------------------
port rule counts
              tcp     udp    icmp      ip
      any       0       0       1       0
    total       0       0       1       0
------------------------------------------------------------
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
```



```
      tcp     udp    icmp      ip
      any       0       0       1       0
    total       0       0       1       0
------------------------------------------------------------
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
10/23-00:29:54.416852 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.129 → 192.168.10.2
10/23-00:29:54.417186 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.2 → 192.168.10.129
10/23-00:29:55.426893 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.129 → 192.168.10.2
10/23-00:29:55.427461 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.2 → 192.168.10.129
10/23-00:29:56.446505 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.129 → 192.168.10.2
10/23-00:29:56.446955 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.2 → 192.168.10.129
10/23-00:29:57.474335 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.129 → 192.168.10.2
10/23-00:29:57.474795 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.2 → 192.168.10.129
10/23-00:30:37.199756 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.129 → 192.168.10.2
10/23-00:30:37.200111 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.2 → 192.168.10.129
10/23-00:30:38.206535 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.129 → 192.168.10.2
10/23-00:30:38.206962 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.2 → 192.168.10.129
10/23-00:30:39.230392 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.129 → 192.168.10.2
10/23-00:30:39.230997 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.2 → 192.168.10.129
10/23-00:30:40.231190 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.129 → 192.168.10.2
10/23-00:30:40.231583 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.10.2 → 192.168.10.129
```